# MEMORANDUM

**TO:** Dr. Tina Han

**FROM:** Hengeltraub, Isabella; Wu, Xintong; Jhanwar, Yuvraj; Hassan, Khaleel

**CC:** ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

**DATE:** March-14-2025

**RE:** Three Point Bending Euler-Bernoulli and Timoshenko Beam Theories: Lab 6

# EXECUTIVE SUMMARY

The primary objective of this lab is to analyze the influence of material and geometric stiffness on the deflection behavior of homogeneous, linear elastic beams which are subjected to three-point bending. Material stiffness is given by the Young's modulus ($E$) of a material while the geometric stiffness is represented by area moment of inertia ($I$) and span length ($L$) of the beam. Young's modulus ($E$) is the mechanical property of a material to withstand compression or elongation with respect to its length [1]. The secondary objective of the lab is to compare how the Euler-Bernoulli and Timoshenko beam theories apply in determining deflection behavior in non-slender, homogeneous, linear elastic beams under three-point bending. The specimen beams were tested using a three-point bending setup, with a concentrated load applied at the center between the two supports 1, as shown below. Prior to the start of the experiment, the beams were measured to get the width,

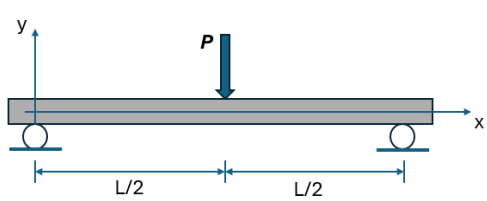



Figure 1: Simplified model of three-point bending

height and length using vernier calipers. Additionally, the beam's material was also taken note of. Table 1 below shows the specimen specifications. The Instron UTM was used to detect and display

the vertical load applied to the beam as they underwent three-point bending. The specimen was secured on the lower anvils, which were positioned at the correct testing distance. The top anvil was lowered carefully until the loading nose was almost in contact with the specimen. Subsequently, the displacement was zeroed and the experiment was started. Upon successful completion of the test, the test data was exported to a secure device for analysis. From the collected data, calculations

| Test | Specimen | Max Load [kN] | L [mm] | Recording Increment [kN] |
|---|---|---|---|---|
| 1 | 1/4 in Aluminum | 0.4 | 200 | 0.05 |
| 2 | 1/2 in Aluminum | 2 | 200 | 0.2 |
| 3 | 3/4 in Aluminum | 2.5 | 200 | 0.25 |
| 4 | 1/4 in 1018 Steel | 0.8 | 200 | 0.08 |
| 5 | 1/4 in Hex Aluminum | 0.1 | 200 | 0.01 |

Table 1: Test Specimen Data

and analyses were performed using MATLAB and the estimated Young's modulus ($E$) for each specimen was given as output. The experimental values were then compared to published values to validate the accuracy and reliability of data and experimental procedures as Table 2 illustrates. All the results were within the 95% confidence intervals. The percent errors were relatively low

| Specimen | $b$ [mm] | $h$ [mm] | $I$ [mm$^4$] | $E$ [GPa] | $CI$ [GPa] | $E_{pub}$ [GPa] | % diff |
|---|---|---|---|---|---|---|---|
| 1/4 in. Al. | 25.40 | 6.35 | 541.97 | 61.36 | [61.33, 61.40] | 69 | 11.1 |
| 1/2 in. Al. | 25.40 | 12.70 | 4335.74 | 51.58 | [51.53, 51.62] | 69 | 25.2 |
| 3/4 in. Al. | 25.40 | 19.05 | 14633.14 | 34.05 | [34.02, 34.08] | 69 | 50.7 |
| 1/4 in. Steel | 25.40 | 6.35 | 541.97 | 156.48 | [156.44, 156.52] | 205 | 23.7 |
| Hex. Al. | 25.40 | 6.35 | 97.7829 | 64.72 | [64.717, 64.722] | 69 | 6.2 |

Table 2: Comparison of experimental and published Young's modulus

except for one specimen with 50.7%. Furthermore, Euler-Bernoulli and Timoshenko equations, denoted by $v_{L/2}^{E-B}$ 2 and $v_{L/2}^{T}$ 3, were used to compute beam deflections and their plots were compared as well. In addition, linear regression was used to estimate the moment of inertia ($I$) of the hexagonal Aluminum. Several potential sources of error existed in this lab. Calibration mistakes in the load cell could cause significant measurement errors, while incorrect dimensions could lead to result discrepancies. Deflection measurement was also prone to error due to signal noise, sensor drift, and misalignment. Lastly, uncertainties in error analysis and propagation further impacted the final results.

# RESULTS

The deflection $D$ of 1/4 inch Aluminum with the load increasing from 0 kN to 0.4 kN in increments of 0.05 kN is shown in the table below:

| Load [kN] | $D$ [mm] |
|---|---|
| 0 | 0 |
| 0.05 | 0.2863 |
| 0.10 | 0.5355 |
| 0.15 | 0.7835 |
| 0.20 | 1.0307 |
| 0.25 | 1.2775 |
| 0.30 | 1.5238 |
| 0.35 | 1.7699 |
| 0.40 | 2.0154 |

Table 3: Measured deflection of 1/4 inch Aluminum

The deflection $D$ of 1/2 inch Aluminum with the load increasing from 0 kN to 2 kN in increments of 0.2 kN is shown in the table below:

| Load [kN] | $D$ [mm] |
|---|---|
| 0 | 0 |
| 0.2 | 0.1865 |
| 0.4 | 0.3358 |
| 0.6 | 0.4825 |
| 0.8 | 0.6280 |
| 1.0 | 0.7726 |
| 1.2 | 0.9167 |
| 1.4 | 1.0602 |
| 1.6 | 1.2030 |
| 1.8 | 1.3453 |
| 2.0 | 1.4874 |

Table 4: Measured deflection of 1/2 inch Aluminum

The deflection $D$ of 3/4 inch Aluminum with the load increasing from 0 kN to 2.5 kN in increments of 0.25 kN is shown in the table below:

| Load [kN] | $D$ [mm] |
|---|---|
| 0 | 0 |
| 0.25 | 0.1245 |
| 0.50 | 0.2107 |
| 0.75 | 0.2956 |
| 1.00 | 0.3795 |
| 1.25 | 0.4624 |
| 1.50 | 0.5446 |
| 1.75 | 0.6260 |
| 2.00 | 0.7069 |
| 2.25 | 0.7875 |
| 2.50 | 0.8680 |

Table 5: Measured deflection of 3/4 inch Aluminum

The deflection $D$ of 1/4 inch 1018 Steel with the load increasing from 0 kN to 0.8 kN in increments of 0.08 kN is shown in the table below:

| Load [kN] | $D$ [mm] |
|---|---|
| 0 | 0 |
| 0.08 | 0.1737 |
| 0.16 | 0.3344 |
| 0.24 | 0.4901 |
| 0.32 | 0.6453 |
| 0.40 | 0.8005 |
| 0.48 | 0.9556 |
| 0.56 | 1.1102 |
| 0.64 | 1.2632 |
| 0.72 | 1.416 |
| 0.80 | 1.5691 |

Table 6: Measured deflection of 1/4 inch 1018 Steel

The deflection $D$ of 1/4 inch hexagonal Aluminum with the load increasing from 0 kN to 0.1 kN in increments of 0.01 kN is shown in the table below:

| Load [kN] | $D$ [mm] |
|---|---|
| 0 | 0 |
| 0.01 | 0.2817 |
| 0.02 | 0.5465 |
| 0.03 | 0.8110 |
| 0.04 | 1.0743 |
| 0.05 | 1.338 |
| 0.06 | 1.6004 |
| 0.07 | 1.8633 |
| 0.08 | 2.1259 |
| 0.09 | 2.3881 |
| 0.10 | 2.6505 |

Table 7: Measured deflection of 1/4 inch hexagonal Aluminum

The Young's modulus of each specimen is estimated using the Euler-Bernoulli theory and linear regression, as seen in the graphs below. Figure 2 shows the plots of the 1/4 inch Steel and Aluminum beams, Figure 3 shows the plots of the 1/4 inch, 1/2 inch, and 3/4 inch Aluminum beams, and Figure 4 shows the plots of the rectangular and hexagonal 1/4 inch Aluminum beams. The linear regression is given by Equation 1, where $\hat{E}$ is the estimated Young's modulus and $\hat{v}_e$ is the error in displacement.

$$P = \frac{48I}{L^3}\hat{E}(v_{L/2} - \hat{v}_e) \tag{1}$$

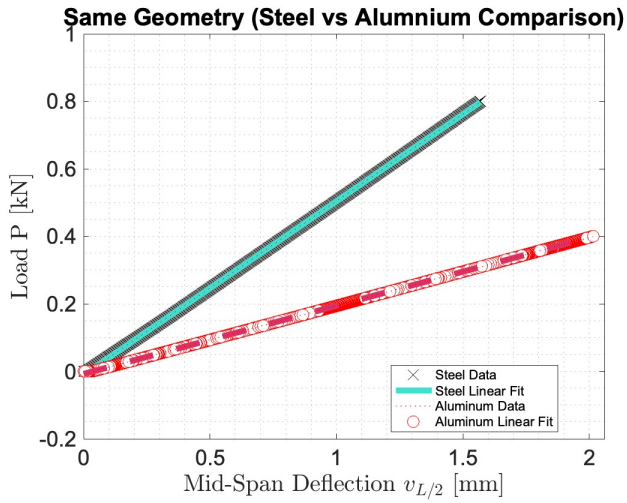



Figure 2: Comparison of 1/4 inch Steel and Aluminum beams

As seen in Figure 2, the slope of Steel is steeper than that of Aluminum, meaning it has a higher Young's modulus.

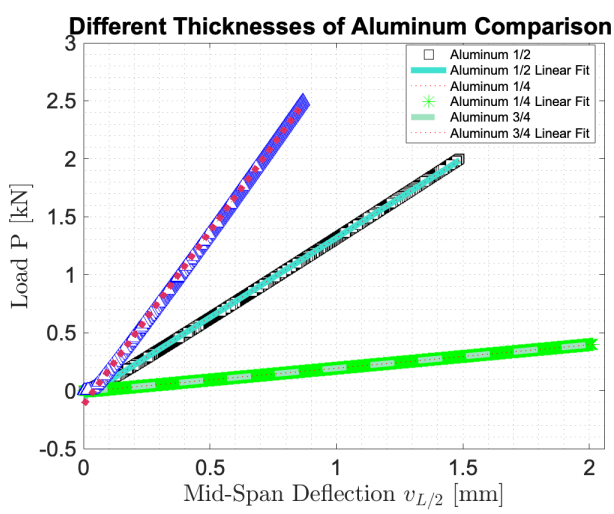



Figure 3: Comparison of 1/4 inch, 1/2 inch, and 3/4 inch Aluminum beams

As seen in Figure 3, higher thickness corresponds to a steeper slope, meaning it has a higher Young's modulus.

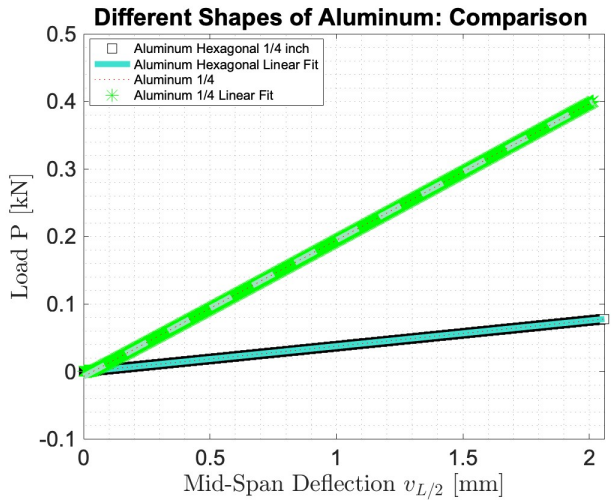



Figure 4: Comparison of 1/4 inch rectangular and hexagonal Aluminum

As seen in Figure 4, the rectangular shape has a steeper slope, meaning it has a higher Young's modulus.

Table 8 shows the experimental Young's moduli and their 95% confidence intervals, as well as compares the experimental Young's moduli with published values [2][3]. $b$ is the length of the base of the specimen, $h$ is the height of the specimen, $I$ is the moment of inertia, $L$ is the length of the specimen, $E$ is the Young's modulus, $CI$ is the 95% confidence interval, $E_{pub}$ is the published Young's modulus, and % diff is the percent difference between the experimental and published Young's modulus.

| Specimen | $b$ [mm] | $h$ [mm] | $I$ [mm$^4$] | $L$ [mm] | $E$ [GPa] | $CI$ [GPa] | $E_{pub}$ [GPa] | % diff |
|---|---|---|---|---|---|---|---|---|
| 1/4 in. Al. | 25.40 | 6.35 | 541.97 | 200 | 61.36 | [61.33, 61.40] | 69 | 11.1 |
| 1/2 in. Al. | 25.40 | 12.70 | 4335.74 | 200 | 51.58 | [51.53, 51.62] | 69 | 25.2 |
| 3/4 in. Al. | 25.40 | 19.05 | 14633.14 | 200 | 34.05 | [34.02, 34.08] | 69 | 50.7 |
| 1/4 in. Steel | 25.40 | 6.35 | 541.97 | 200 | 156.48 | [156.44, 156.52] | 205 | 23.7 |
| Hex. Al. | 25.40 | 6.35 | 97.7829 | 200 | 64.72 | [64.717, 64.722] | 69 | 6.2 |

Table 8: Comparison of experimental and published Young's modulus

Linear regression is once again used, alongside the estimated Young's modulus in Table 8, to estimate the moment of inertia of the hexagonal Aluminum. Table 9 compares it with the theoretical value and displays its 95% confidence interval, with $CI_I$ being the 95% confidence interval for the moment of inertia and $I_{theo}$ being the theoretical value.

| Specimen | $E$ [GPa] | $CI_E$ [GPa] | $E_{pub}$ [GPa] | $I$ [mm$^4$] | $CI_I$ [mm$^4$] | $I_{theo}$ [mm$^4$] | % diff |
|---|---|---|---|---|---|---|---|
| Hex. Al. | 64.72 | [64.717, 64.722] | 69 | 97.78 | [97.78, 97.79] | 97.78 | 0 |

Table 9: Comparison of experimental and theoretical moment of inertia of hexagonal specimen

The Euler-Bernoulli and Timoshenko beam theories are compared in the graphs below by plotting the deflection and load of the Euler-Bernoulli and Timoshenko beam theories for the 1/4 inch, 1/2 inch, and 3/4 inch Aluminum beams. The Euler-Bernoulli equation is given in Equation 2, while the Timoshenko equation is given in Equation 3, where $E$ is the Young's modulus, $G$ is the shear modulus, $A$ is the area of the cross-section, and $K_s$ is the Timoshenko shear correction factor.

$$v_{L/2}^{E-B} = \frac{PL^3}{48EI} \tag{2}$$

$$v_{L/2}^{T} = \frac{PL^3}{48EI} + \frac{PL}{4K_sGA} \tag{3}$$

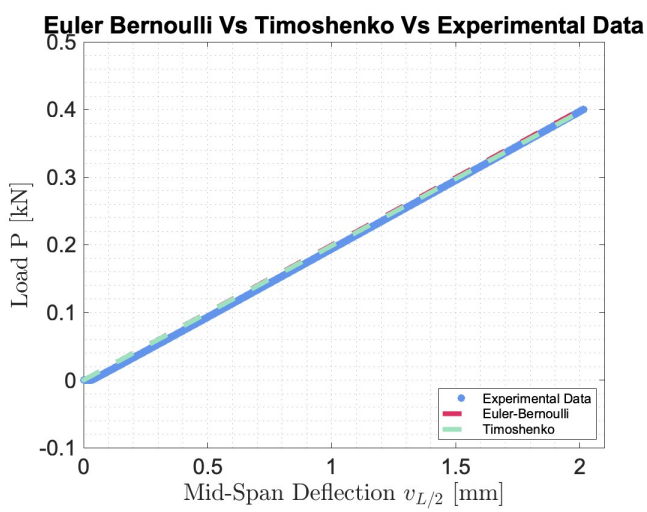



Figure 5: Comparison of beam theories for 1/4 inch Aluminum

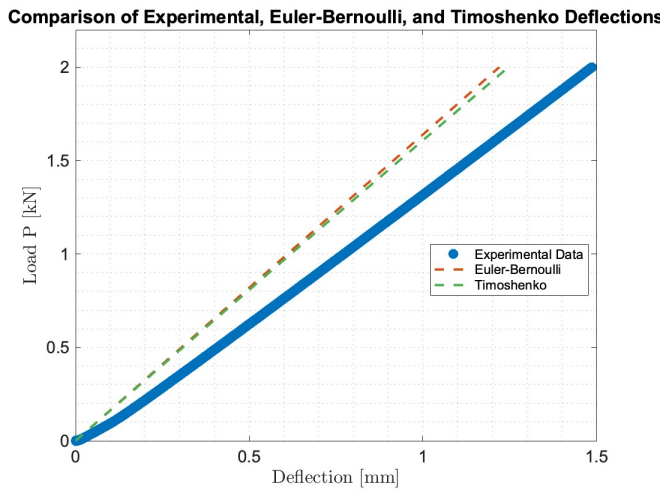



Figure 6: Comparison of beam theories for 1/2 inch Aluminum

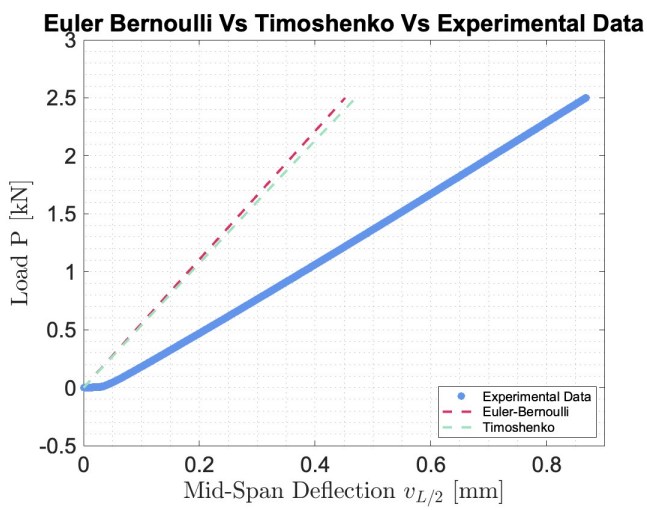



Figure 7: Comparison of beam theories for 3/4 inch Aluminum

As can be seen in Figures 5, 6, and 7, the beam theories are closer to the experimental data when the thickness is smaller.

The relative difference between the Euler-Bernoulli and Timoshenko beam theories are plotted for different values of Poisson's ratio in Figure 8, showing a higher difference for the higher Poisson's ratio. The equation for relative difference is derived below, with the assumptions that $K_s = \frac{5+5\nu}{6+5\nu}$, $G = \frac{E}{2(1+\nu)}$, $A = bh$, and $I = \frac{1}{12}bh^3$.

$$
\begin{aligned}
\frac{v_{L/2}^{T} - v_{L/2}^{E-B}}{v_{L/2}^{E-B}} &= \frac{\frac{PL^3}{48EI} + \frac{PL}{4K_sGA} - \frac{PL^3}{48EI}}{\frac{PL^3}{48EI}} \\
&= \frac{PL}{4K_sGA} * \frac{48EI}{PL^3} \\
&= \frac{12EI}{K_sGAL^2} \\
&= \frac{12Ebh^3}{12K_sGAL^2} \\
&= \frac{2Ebh^3(6+5\nu)(1+\nu)}{(5+5\nu)EL^2A} \\
&= \frac{2bh^3(6+5\nu)}{5L^2bh} \\
&= \frac{2}{5}(6+5\nu)(L/h)^{-2}
\end{aligned}
\tag{4}
$$

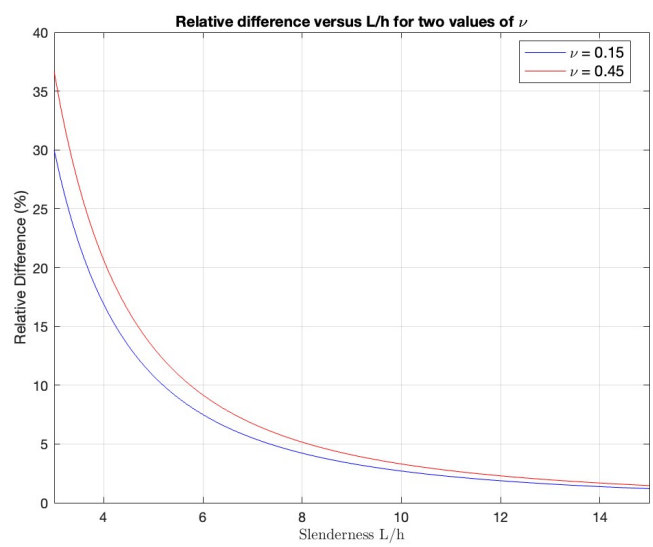



Figure 8: Relative difference between the Euler-Bernoulli and Timoshenko beam theories

# CONCLUSIONS

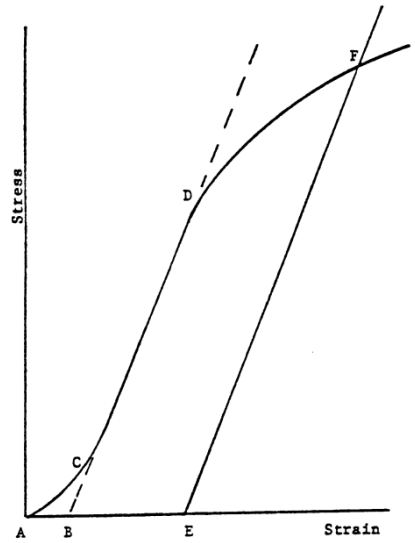
There were two primary objectives of this lab, the first being: to analyze the influence of material and geometric stiffness on the deflection behavior of homogeneous, linear elastic beams that are subjected to three-point bending. The secondary objective of the lab is to compare how the Euler-

Bernoulli and Timoshenko beam theories apply in determining deflection behavior in non-slender, homogeneous, linear elastic beams under three-point bending.

The empirical values for Young's Modulus provided mixed conclusions in proving the validity of the three-point bending formula for deflection. The lowest percent difference between experimental and published Young's modulus was for the hexagonal aluminum beam at 6.2 percent error (PE). The second lowest was the 1/4 inch aluminum beam at 11.1 PE, followed by the 1/4 inch steel at 23.7 PE, followed by the 1/2 inch aluminum with 25.2 PE, with the highest percent error being for the 3/4 inch aluminum beam at 50.7% —all of which were within their respective 95% confidence value ranges. Further, as can be seen in Figure 5, the 1/4 inch aluminum beam theories and experimental data align very closely; however, as can be seen in Figure 6, both theories misaligned with the 1/2 inch aluminum beam experimental data, with greater misalignment for higher loads, and even higher misalignment in Figure 7, for the comparison of beam theories for 3/4 inch aluminum beam. Both these percent errors and graphical analysis prompt the conclusion that the beam theories best align with the experimental data for smaller beam thicknesses. Additionally, as can be seen in Figure 8, there are also differences within Euler-Bernoulli and Timoshenko beam theories for different Poisson's ratios, with smaller differences between the therories for smaller Poisson's ratios. Thus, the three-point bending formula for deflection seems to be valid for thinner beams, as well as lower loads, with best alignment between theories for lower poissant ratios.

On the other hand, the experimental values for moment of inertial fully verified the formula for the deflection of a simply-supported beam under three point bending. The published value for hexagonal aluminum was 97.78 mm cubed, which was the exact same value found in this experiment, with a 0% relative difference between the values.

As for the actual processing of experimental data, the ASTM Standard D-790 (annex A1) procedure advises removal of the toe region, segment AC, as can be seen in Figure 9. This section is caused by a "take-up of slack and alignment or seating of the specimen" [4]. It interferes with the correct calculation of material properties, such as Young's modulus, which was calculated in this lab. In




NOTE—Some chart recorders plot the mirror image of this graph.
**FIG. A1.1 Material with Hookean Region**

Figure 9: ASTM Standard D-790 (Annex A1)

cases such as this lab where the data has a linear portion, in order to account for this irregularity, section AC should be removed, and a line should be extended out from section CD into a new zero-strain point at B. However, the team for this lab did not remove the toe region for the lab's data, as this region did not appear to be present. One potential reason why this toe region was avoided is that the calibration was completed prior to the start of the experiment. For this calibration, the Instron UT Machine's top anvil was slowly lowered until making contact with the specimen, at which point the team noted a force spike in the machine. This anvil was then raised until that force returned to zero, at which point the displacement was manually set to zero, which could have had the same effect as the creation of point B in Figure 9, where a new zero strain was found.

It is also recommended by the ASTM Standard D-790, section 4.1, that a bar with a rectangular cross section with a loading halfway between supports (which was the setup used in this lab) should have a 16:1 span-to-depth ratio - span being the distance between supports and depth meaning the depth of the specimen [4]. Following this allows the prevention of errors that might be caused by shear forces or potentially shear deformation of the material, as it is pushed down by a load. The

speciments with their respective span, depth, and the span-depth ratios can be seen in Table 10.

| Specimen | Span [mm] | Depth [mm] | Ratio |
|---|---|---|---|
| 1/4 in. Al. | 200 | 6.35 | 31.50 |
| 1/2 in. Al. | 200 | 12.70 | 15.75 |
| 3/4 in. Al. | 200 | 19.05 | 10.49 |
| 1/4 in. Steel | 200 | 6.35 | 31.50 |
| Hex. Al. | 200 | 6.35 | 31.50 |

Table 10: Span-Depth Table

It is worthy to note that the three specimens with the highest ratio of 31.50 had the lowest percent difference between experimental and published Young's modulus as seen by Table 8, while the specimen with the lowest ratio of 10.49, which is significantly below the suggested 16:1 span-depth ratio, had the highest percent difference of 50.7% between experimental and published Young's modulus. Meanwhile, the 1/2 inch aluminum sample, which was very slightly below 16:1, with a 15.75 ratio, had a lower percentage difference of 25. 2%, but this PE was still higher than the samples that were above that ratio. This leads to the conclusion that following the 16:1 span-depth in the testing of beams allows for more accurate testing data and lower percentage errors, as well as that not having followed this ratio for the beams in this lab could have been a major source of error.

# REFERENCE LIST

[1] BYJU'S. "Modulus of elasticity - definition, young's modulus, formula, unit, faqs." (2023), [Online]. Available: `https://byjus.com/physics/youngs-modulus-elastic-modulus/` (visited on 03/14/2025).

[2] MakeItFrom. "6061-t6511 aluminum." (2020), [Online]. Available: `https://www.makeitfrom.com/material-properties/6061-T6511-Aluminum` (visited on 03/08/2025).

[3] A. Materials. "Aisi 1018 carbon steel (uns g10180)." (2013), [Online]. Available: `https://www.azom.com/article.aspx?ArticleID=9138` (visited on 03/08/2025).

[4] ASTM. "Standard test methods for flexural properties of unreinforced and reinforced plastics and electrical insulating material." (2016), [Online]. Available: https://spada.uns.ac.id/pluginfile.php/195209/mod_resource/content/1/D%20790.%28three%20point%20bend%29.pdf.

# DATA SHEETS

| | A | B | C |
|---|---|---|---|
| 1 | Time | Displacement | Force |
| 2 | (s) | (mm) | (N) |
| 3 | 0 | 0 | -0.0043 |
| 4 | 0.02 | 0 | 0.0166 |
| 5 | 0.04 | 0 | 0.0207 |
| 6 | 0.06 | 0 | -0.0023 |
| 7 | 0.08 | 0 | 0.0019 |
| 8 | 0.1 | 0 | 0.0072 |
| 9 | 0.12 | 0 | -0.0088 |
| 10 | 0.14 | 0 | -0.0143 |
| 11 | 0.16 | 0.0001 | 0.0081 |
| 12 | 0.18 | 0.0001 | 0.0116 |
| 13 | 0.2 | 0.0001 | 0.0095 |
| 14 | 0.22 | 0.0001 | 0.0012 |
| 15 | 0.24 | 0.0001 | -0.0074 |
| 16 | 0.26 | 0.0002 | -0.0104 |
| 17 | 0.28 | 0.0002 | -0.0117 |
| 18 | 0.3 | 0.0002 | 0.0005 |
| 19 | 0.32 | 0.0002 | 0.004 |
| 20 | 0.34 | 0.0003 | 0.0059 |
| 21 | 0.36 | 0.0003 | 0.0002 |
| 22 | 0.38 | 0.0003 | 0.0113 |
| 23 | 0.4 | 0.0004 | 0.0107 |
| 24 | 0.42 | 0.0004 | 0.002 |
| 25 | 0.44 | 0.0005 | 0.005 |
| 26 | 0.46 | 0.0005 | -0.0017 |
| 27 | 0.48 | 0.0006 | -0.0106 |
| 28 | 0.5 | 0.0007 | -0.0059 |
| 29 | 0.52 | 0.0007 | 0.0101 |
| 30 | 0.54 | 0.0008 | 0.0151 |
| 31 | 0.56 | 0.0008 | 0.017 |
| 32 | 0.58 | 0.0009 | 0.0056 |

Figure 10:
Displacement and Force Data for 1/4 in Diameter for Rectangular Aluminum

| | A | B | C |
|---|---|---|---|
| 1 | Time | Displacement | Force |
| 2 | (s) | (mm) | (N) |
| 3 | 0 | 0 | -0.0061 |
| 4 | 0.02 | 0 | -0.0076 |
| 5 | 0.04 | 0 | 0.005 |
| 6 | 0.06 | 0 | 0.0092 |
| 7 | 0.08 | 0 | 0.0034 |
| 8 | 0.1 | 0 | -0.0113 |
| 9 | 0.12 | 0 | -0.0007 |
| 10 | 0.14 | 0 | 0.0017 |
| 11 | 0.16 | 0.0001 | 0.0131 |
| 12 | 0.18 | 0.0001 | 0.0182 |
| 13 | 0.2 | 0.0001 | 0.0225 |
| 14 | 0.22 | 0.0001 | 0.0247 |
| 15 | 0.24 | 0.0001 | 0.0253 |
| 16 | 0.26 | 0.0001 | 0.0225 |
| 17 | 0.28 | 0.0002 | 0.0267 |
| 18 | 0.3 | 0.0002 | 0.0393 |
| 19 | 0.32 | 0.0002 | 0.0057 |
| 20 | 0.34 | 0.0002 | -0.0214 |
| 21 | 0.36 | 0.0003 | -0.0266 |
| 22 | 0.38 | 0.0003 | -0.0363 |
| 23 | 0.4 | 0.0003 | -0.0238 |
| 24 | 0.42 | 0.0003 | -0.0154 |
| 25 | 0.44 | 0.0004 | -0.015 |
| 26 | 0.46 | 0.0004 | -0.0206 |
| 27 | 0.48 | 0.0004 | -0.0192 |
| 28 | 0.5 | 0.0005 | -0.0102 |
| 29 | 0.52 | 0.0005 | -0.0031 |
| 30 | 0.54 | 0.0005 | 0.0074 |
| 31 | 0.56 | 0.0006 | 0.0069 |
| 32 | 0.58 | 0.0006 | 0.0053 |

Figure 11:
Displacement and Force Data for 1/2 in Diameter for Rectangular Aluminum

| | A | B | C |
|---|---|---|---|
| 1 | Time | Displacement | Force |
| 2 | (s) | (mm) | (N) |
| 3 | 0 | 0 | 0.0004 |
| 4 | 0.02 | 0 | 0.0087 |
| 5 | 0.04 | 0 | 0.0052 |
| 6 | 0.06 | 0 | 0.0073 |
| 7 | 0.08 | 0 | 0.0002 |
| 8 | 0.1 | 0 | -0.0073 |
| 9 | 0.12 | 0 | 0.0042 |
| 10 | 0.14 | 0 | -0.005 |
| 11 | 0.16 | 0 | 0.0029 |
| 12 | 0.18 | 0.0001 | -0.0053 |
| 13 | 0.2 | 0.0001 | 0.0001 |
| 14 | 0.22 | 0.0001 | 0.02 |
| 15 | 0.24 | 0.0001 | 0.0292 |
| 16 | 0.26 | 0.0001 | 0.0225 |
| 17 | 0.28 | 0.0002 | 0.0086 |
| 18 | 0.3 | 0.0002 | 0.0101 |
| 19 | 0.32 | 0.0002 | 0.0101 |
| 20 | 0.34 | 0.0002 | 0.0115 |
| 21 | 0.36 | 0.0003 | 0.0045 |
| 22 | 0.38 | 0.0003 | -0.0037 |
| 23 | 0.4 | 0.0003 | -0.0005 |
| 24 | 0.42 | 0.0003 | -0.0026 |
| 25 | 0.44 | 0.0004 | 0.0009 |
| 26 | 0.46 | 0.0004 | 0.0189 |
| 27 | 0.48 | 0.0004 | 0.0069 |
| 28 | 0.5 | 0.0005 | -0.0007 |
| 29 | 0.52 | 0.0005 | -0.0016 |
| 30 | 0.54 | 0.0005 | 0.0075 |
| 31 | 0.56 | 0.0006 | 0.0082 |
| 32 | 0.58 | 0.0006 | 0.0195 |

Displacement and Force Data for 1/4 in Diameter for Rectangular Steel

| | A | B | C |
|---|---|---|---|
| 1 | Time | Displacement | Force |
| 2 | (s) | (mm) | (N) |
| 3 | 0 | 0 | 0.0035 |
| 4 | 0.02 | 0 | 0.0008 |
| 5 | 0.04 | 0 | 0 |
| 6 | 0.06 | 0 | 0.0123 |
| 7 | 0.08 | 0 | 0.0162 |
| 8 | 0.1 | 0 | 0.0166 |
| 9 | 0.12 | 0 | 0.0135 |
| 10 | 0.14 | 0 | 0.0001 |
| 11 | 0.16 | 0 | 0.0104 |
| 12 | 0.18 | 0.0001 | -0.0008 |
| 13 | 0.2 | 0.0001 | 0.014 |
| 14 | 0.22 | 0.0001 | 0.0082 |
| 15 | 0.24 | 0.0001 | 0.0066 |
| 16 | 0.26 | 0.0001 | 0.0052 |
| 17 | 0.28 | 0.0001 | 0.0255 |
| 18 | 0.3 | 0.0001 | 0.0077 |
| 19 | 0.32 | 0.0001 | 0.0092 |
| 20 | 0.34 | 0.0001 | 0.0159 |
| 21 | 0.36 | 0.0002 | 0.0039 |
| 22 | 0.38 | 0.0002 | -0.003 |
| 23 | 0.4 | 0.0002 | 0.0018 |
| 24 | 0.42 | 0.0002 | -0.0011 |
| 25 | 0.44 | 0.0002 | 0.0082 |
| 26 | 0.46 | 0.0002 | 0.0078 |
| 27 | 0.48 | 0.0003 | 0.0081 |
| 28 | 0.5 | 0.0003 | 0.0009 |
| 29 | 0.52 | 0.0003 | -0.0073 |
| 30 | 0.54 | 0.0003 | -0.0085 |
| 31 | 0.56 | 0.0004 | 0.0002 |
| 32 | 0.58 | 0.0004 | -0.0035 |

Displacement and Force Data for 1/4 in Diameter for Hexagonal Aluminum

| | Time | Displacemen | Force |
|---|---|---|---|
| 2 | | | |
| 3 | (s) | (mm) | (N) |
| 4 | 0 | 0 | -0.0472 |
| 5 | 0.02 | 0 | -0.041 |
| 6 | 0.04 | 0 | -0.049 |
| 7 | 0.06 | 0 | -0.045 |
| 8 | 0.08 | 0 | -0.0423 |
| 9 | 0.1 | 0 | -0.0434 |
| 10 | 0.12 | 0 | -0.0261 |
| 11 | 0.14 | 0 | -0.0429 |
| 12 | 0.16 | 0.0001 | -0.0674 |
| 13 | 0.18 | 0.0001 | -0.0491 |
| 14 | 0.2 | 0.0001 | -0.042 |
| 15 | 0.22 | 0.0001 | -0.043 |
| 16 | 0.24 | 0.0001 | -0.0444 |
| 17 | 0.26 | 0.0001 | -0.0563 |
| 18 | 0.28 | 0.0001 | -0.0432 |
| 19 | 0.3 | 0.0001 | -0.0501 |
| 20 | 0.32 | 0.0002 | -0.0615 |
| 21 | 0.34 | 0.0002 | -0.0458 |
| 22 | 0.36 | 0.0002 | -0.0474 |
| 23 | 0.38 | 0.0002 | -0.0448 |
| 24 | 0.4 | 0.0003 | -0.0254 |
| 25 | 0.42 | 0.0003 | -0.0352 |
| 26 | 0.44 | 0.0003 | -0.0234 |
| 27 | 0.46 | 0.0004 | -0.0309 |
| | 0.48 | 0.0004 | -0.0532 |

Figure 12:
Displacement and Force Data for 3/4 in Diameter for Rectangular Aluminum

# APPENDIX

The following code was used to calculate values of E (Young's Modulus) and the $\nu$ (deflection) for the Aluminum beam with 0.75 in x 1 in cross-sectional area. The code also produced the 95% Confidence Intervals for both E and $\nu$. The same code was edited four more times to calculate the same two parameters, E and $\nu$ for the other two Aluminum beams with 0.25 in X 1 in and 0.5 in X 1 in cross-sections, the steel beam, and the Aluminum hexagonal beam. The editing was primarily changing the readmatrix command and the I values.

```
function three_point_bending_model()
    % In the space below, import the data sheet(s) obtained from the Instron
   UTM.


    Alumdata = readmatrix("Al_075_Lab6_Test1 (1).csv");


    % Referencing the imported data sheet, define the list of values for
    % vertical deflection and force applied.
    v_t = Alumdata(1:8682,2); % Vertical Deflection [mm]
    P_t = Alumdata(1:8682,3)/1000; % Applied Force [kN]
    % Calculate the area moment of intertia and define the length of
    % the specimen.
    I = ((19.05)^3*(25.4))/12; % Area moment of intertia [mm^4]
    L = 200; % Length of the beam [mm]
    % Defining the number of data points and number of bootstrap samples
    n = length(P_t); % number of data points
    numBootstraps = 1000; % number of bootstrap samples
    % Initial guess for 'Youngs modulus E and error in displacement vE
    E0 = 1;
    vE0 = 0;
    % Define objective function
```

```
objfun = @(x) sum((P_t - (48*I*x(1)/L^3) .* (v_t-x(2))).^2);
% Define objective function for bootstrapping
objfun_bootstrap = @(x,P_t,I,L,v_t) sum((P_t - (48*I*x(1)/L^3) .* (v_t-x
(2))).^2);
% Define constraints for E (E > 0)
options = optimoptions(@fmincon,'Algorithm','interior-point');
lb = [0, -Inf]; % lower bound
ub = [Inf, Inf]; % upper bound
% Call fmincon to minimize objective function
x = fmincon(objfun,[E0, vE0],[],[],[],[],lb,ub,[],options);
% Initialize array to hold bootstrap estimates
bootstrap_estimates = zeros(numBootstraps,2);
% Perform bootstrap estimation
for i = 1:numBootstraps
    % Generate bootstrap sample
    ind = randi(n,n,1);
    P_b = P_t(ind);
    I_b = I;
    L_b = L;
    v_b = v_t(ind);
    % Call fmincon to minimize objective function for bootstrap sample
    x_b = fmincon(@(x) objfun_bootstrap(x,P_b,I_b,L_b,v_b),...
    [E0,vE0],[],[],[],[],lb,ub,[],options);
    bootstrap_estimates(i,:) = x_b;
end
% Compute confidence intervals
alpha = 0.05; % 95% confidence interval
E_conf_int = quantile(bootstrap_estimates(:,1),[alpha/2, 1-alpha/2]);
vE_conf_int = quantile(bootstrap_estimates(:,2),[alpha/2, 1-alpha/2]);
% Display the optimal value of E and vE
disp(['Optimal value of E: ', num2str(x(1)), ' GPa']);
disp(['Optimal value of vE: ', num2str(x(2)), ' mm']);
% Display the confidence intervals
```

```
    disp(['95% confidence interval for E: [', num2str(E_conf_int(1)), ', ', num2str(E_conf_int(2)), ']', ' GPa'])
    disp(['95% confidence interval for vE: [', num2str(vE_conf_int(1)), ', ', num2str(vE_conf_int(2)), ']', ' mm'])
end
```

The following code was used to produce plots of Load P vs deflection at mid span.

```
%%%
% Do not change, this list includes 9 preset colors
% Feel free to use these colors in your plot
colorlist=["#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f"];

%%%
%Edit this portion, Use readmatrix function to import axial and lateral
%strain data

data1 = readmatrix("2025_SP_Section08_Team04_Lab06_HEX_2_18.xlsx");
data2 = readmatrix("2025_SP_Section08_Team04_Lab06_Al14_22.xlsx");
%data3= readmatrix("Al_075_Lab6_Test1 (1).csv");

% Extract data (Steel)
 v_hex = data1(1:end,2); % Vertical Deflection [mm]
 P_hex = data1(1:end,3)/1000; % Applied Force [kN]

% Extract data (Aluminum)
v_f = data2(1:end,2); % Vertical Deflection [mm] (Aluminum)
P_f = data2(1:end,3)/1000; % Applied Force [kN] (Aluminum)

% v_t = data3(1:end,2); % Vertical Deflection [mm]
%  P_t = data3(1:end,3)/1000; % Applied Force [kN]
%
% %%%
```

```
%Do not change, mdl is the linear regression model for steel
mdl_hex = fitlm(v_hex, P_hex, 'VarNames', {'v_hex', 'P_hex'});

%%% Linear regression model for Aluminum data
mdl_f = fitlm(v_f, P_f, 'VarNames', {'v_f', 'P_f'});

%mdl_hex = fitlm(v_hex, P_hex, 'VarNames', {'v_hex', 'P_hex'});

% mdl_t = fitlm(v_t, P_t, 'VarNames', {'v_t', 'P_t'});

%%%
% Plot experimental data and regression for Steel and Aluminum
figure(1)
hold on

mdl_plot_hex = plot(mdl_hex);
delete(mdl_plot_hex(3)); % Remove unnecessary plot component
mdl_plot_hex(1).MarkerSize = 10;
mdl_plot_hex(1).Marker = "s"; % Square marker
mdl_plot_hex(1).Color = "k";
mdl_plot_hex(1).MarkerFaceColor = "auto";
mdl_plot_hex(2).LineStyle = "-"; % Solid line for Steel fit
mdl_plot_hex(2).LineWidth = 5;
mdl_plot_hex(2).Color = colorlist(4); % Unique color for Steel fit

% Aluminum Data (Red Diamond markers)
mdl_plot_f = plot(mdl_f);
delete(mdl_plot_f(3)); % Remove unnecessary plot component
mdl_plot_f(1).MarkerSize = 10;
mdl_plot_f(1).Marker = "*"; % Diamond marker
mdl_plot_f(1).Color = "g";
mdl_plot_f(1).MarkerFaceColor = "auto";
mdl_plot_f(2).LineStyle = "--"; % Dashed line for Aluminum fit
```

```
mdl_plot_f(2).LineWidth = 5;
mdl_plot_f(2).Color = colorlist(5); % Unique color for Aluminum fit

%
% % Titanium Data (Blue Triangle markers)
% mdl_plot_hex = plot(mdl_hex);
% delete(mdl_plot_hex(3)); % Remove unnecessary plot component
% mdl_plot_hex(1).MarkerSize = 10;
% mdl_plot_hex(1).Marker = "^"; % Triangle marker
% mdl_plot_hex(1).Color = "b";
% mdl_plot_hex(1).MarkerFaceColor = "auto";
% mdl_plot_hex(2).LineStyle = ":"; % Dotted line for Titanium fit
% mdl_plot_hex(2).LineWidth = 5;
% mdl_plot_hex(2).Color = colorlist(6); % Unique color for Titanium fit

% % Titanium Data (Blue Triangle markers)
% mdl_plot_t = plot(mdl_t);
% delete(mdl_plot_t(3)); % Remove unnecessary plot component
% mdl_plot_t(1).MarkerSize = 10;
% mdl_plot_t(1).Marker = "^"; % Triangle marker
% mdl_plot_t(1).Color = "b";
% mdl_plot_t(1).MarkerFaceColor = "auto";
% mdl_plot_t(2).LineStyle = ":"; % Dotted line for Titanium fit
% mdl_plot_t(2).LineWidth = 5;
% mdl_plot_t(2).Color = colorlist(6); % Unique color for Titanium fit


%%%
% Add linear fit equations for Steel and Aluminum
Eq_text_hex = sprintf('$\\varepsilon_L^{(s)} = %.4f \\varepsilon_A + %.4f\\
    times 10^{4}$' ...
        , mdl_hex.Coefficients.Estimate(2), mdl_hex.Coefficients.Estimate(1) *
     10000);
```

```
Eq_Anno_hex = text(0.2, -1.85, Eq_text_hex, 'Interpreter', 'latex', 'FontName', 'Arial', 'FontSize', 18, 'Color', 'k');

Eq_text_f = sprintf('$\\varepsilon_L^{(f)} = %.4f \\varepsilon_A + %.4f\\times 10^{4}$' ...
        , mdl_f.Coefficients.Estimate(2), mdl_f.Coefficients.Estimate(1) * 10000);
Eq_Anno_f = text(0.2, -2.25, Eq_text_f, 'Interpreter', 'latex', 'FontName', 'Arial', 'FontSize', 18, 'Color', 'r');
% Eq_text_t = sprintf('$\\varepsilon_L^{(f)} = %.4f \\varepsilon_A + %.4f\\times 10^{4}$' ...
%         , mdl_t.Coefficients.Estimate(2), mdl_t.Coefficients.Estimate(1) * 10000);
% Eq_Anno_t = text(0.2, -2.25, Eq_text_t, 'Interpreter', 'latex', 'FontName', 'Arial', 'FontSize', 18, 'Color', 'r');

%%%
%Do not change this section
box on
grid minor
title('');

%%%
% Edit this section to include proper axis label and legend
xlabel('Mid-Span Deflection $v_{L/2}$ [mm]', 'interpreter', 'latex');
ylabel('Load P [kN]', 'interpreter', 'latex');
title("Different Shapes of Aluminum: Comparison ")
legend({'Aluminum Hexagonal 1/4 inch', 'Aluminum Hexagonal Linear Fit', 'Aluminum 1/4', 'Aluminum 1/4 Linear Fit'}, 'fontsize', 11, 'Location', 'best')
xlim([0 2.06])
%%%
% Formatting: Font size and font name
```

```
ax = gca;
ax.FontName = 'Arial';
ax.FontSize = 18;
```

The following code was used to plot the comparison graphs for the experimental data versus the two theories: Euler Bernoulli and Timoshenko. This code was modified and run 3 times for the three different sizes of Aluminum.

```
%%%
% Define Colors (Light colors for better visibility)
colorlist = {"#0072BD", "#D95319", "#4CAF50"}; % Blue (Exp), Red (EB), Green (
    TEM)

%%%
% Read Data
data1 = readmatrix("2024_Fall_Section08_Team04_Lab6_ALhalf_is_flex_34.xlsx");

% Extract Data
v_h = data1(:,2); % Vertical Deflection [mm] (Experimental)
P_h = data1(:,3)/1000; % Applied Force [kN] (Experimental)

% Calculations
k_s = (5 + (5 .* v_h)) ./ (6 + (5 .* v_h));
G = 63 ./ (2 * (1 + v_h));

v_eb_h = (P_h .* (200)^3) ./ (48 * (((12.7)^3*(25.4))/12) * 63); % Euler-
    Bernoulli
v_tem_h = v_eb_h + (((P_h * 200) ./ ((4 * 322.58) .* k_s .* G))); % Timoshenko

%%%
% Plot all three v terms with P_h
figure(1)
hold on
```

```
% **Experimental Data (Blue Circles)**
plot(v_h, P_h, 'o-', 'MarkerSize', 6, 'Color', colorlist{1}, 'MarkerFaceColor'
    , colorlist{1});

% **Euler-Bernoulli (Red Squares)**
plot(v_eb_h, P_h,'r*', 'MarkerSize', 6, 'Color', colorlist{2}, '
    MarkerFaceColor', colorlist{2});

% **Timoshenko (Green Diamonds)**
plot(v_tem_h, P_h, 'd', 'MarkerSize', 6, 'Color', colorlist{3}, '
    MarkerFaceColor', colorlist{3});

%%%
% Formatting and Labels
xlabel('Deflection [mm]', 'Interpreter', 'latex');
ylabel('Load P [kN]', 'Interpreter', 'latex');
title("Comparison of Experimental, Euler-Bernoulli, and Timoshenko Deflections
    ")

% Custom Legend
legend({'Experimental Data', 'Euler-Bernoulli', 'Timoshenko'}, ...
    'FontSize', 11, 'Location', 'best')

xlim([0 1.5]) % Match the x-axis range in the image
ylim([0 max(P_h) * 1.1]) % Scale to max value of P_h

% Formatting
ax = gca;
ax.FontName = 'Arial';
ax.FontSize = 14;
grid minor
box on
```

```
hold off
```

The following code was used to plot the Relative percentage difference of two different poisson's ratios, 0.15 and 0.45, versus slenderness (L/h).

```
%%%
% Do not change, this list includes 9 preset colors
% Feel free to use these colors in your plot
colorlist = {"#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f"};

%%%
% Read data
data2 = readmatrix("2024_Fall_Section08_Team04_Lab6_ALhalf_is_flex_34.xlsx");

% Extract data (Aluminum)
v_t = data2(:,2); % Vertical Deflection [mm] (Aluminum)
P_t = data2(:,3)/1000; % Applied Force [kN] (Aluminum)

% Calculation
k_s = (5 + 5 .* v_t) ./ (6 + 6 .* v_t);
G = 63 ./ (2 * (1 + v_t));
v_eb = (P_t .* (200)^3) ./ (48 * (((19.05)^3*(25.4))/12) * 63);
v_tem = (P_t .* (200)^3) ./ (48 * (((19.05)^3*(25.4))/12 * 63)) + ((P_t * 200)
    ./ ((4 * 483.87) .* k_s .* G));
L_h = linspace(3, 15, 100);
real_diff_v1 = ((2/5).*(6+5.*(0.15)).*(L_h).^-2).*100;
real_diff_v2 = ((2/5).*(6+5.*(0.45)).*(L_h).^-2).*100;
figure(1)
plot(L_h, real_diff_v1,'-b', 'MarkerSize', 6);
hold on
plot(L_h, real_diff_v2, 'r-', 'MarkerSize', 6);
% Formatting and labels
```

```
xlabel('Slenderness L/h', 'Interpreter', 'latex');
ylabel('Relative Difference (%)', 'Interpreter', 'latex');
title("Relative difference versus L/h for two values of \nu")
grid on


% Custom Legend
legend({'\nu = 0.15', '\nu = 0.45'}, ...
    'FontSize', 11, 'Location', 'best')

%
% %%%
% % Plot all three deflections against P_f
% figure(1)
% hold on
%
% % **Experimental Data (Blue Circles)**
% plot(v_t, P_t, 'o', 'MarkerSize', 6, 'Color', colorlist{3}, 'MarkerFaceColor', colorlist{3});
%
% % **Euler-Bernoulli (Red Stars)**
% plot(v_eb, P_t, '*', 'MarkerSize', 8, 'Color', colorlist{6}); % Red Stars
%
% % **Timoshenko (Green Diamonds)**
% plot(v_tem, P_t, 'd', 'MarkerSize', 6, 'Color', colorlist{5}, 'MarkerFaceColor', colorlist{5});
%
% %%%
% % Formatting and labels
% xlabel('Mid-Span Deflection $v_{L/2}$ [mm]', 'Interpreter', 'latex');
% ylabel('Load P [kN]', 'Interpreter', 'latex');
% title("Euler Bernoulli Vs Timoshenko Vs Experimental Data")
%
```

```
% % Custom Legend
% legend({'Experimental Data', 'Euler-Bernoulli', 'Timoshenko'}, ...
%     'FontSize', 11, 'Location', 'best')
%
xlim([3 15])
%
% % Formatting
% ax = gca;
% ax.FontName = 'Arial';
% ax.FontSize = 18;
% box on
% grid minor
% hold off
```

The following code was used to calculate the I value and its 95% Confidence Intervals based on the calculated E value of the hexagonal Aluminum beam.

```
function three_point_bending_model()
    % In the space below, import the data sheet(s) obtained from the Instron
   UTM.

    Alumdata_two = readmatrix("2025_SP_Section08_Team04_Lab06_HEX_2_18.xlsx");

    % Referencing the imported data sheet, define the list of values for
    % vertical deflection and force applied.
    v_hex = Alumdata_two(1:end,2); % Vertical Deflection [mm]
    P_hex = Alumdata_two(1:end,3)/1000; % Applied Force [kN]

    % Define the length of the beam
    L = 200; % Length of the beam [mm]

    % Given Young's modulus E
    E = 64.7197; % GPa
```

```
% Initial guess for I and error in displacement vE
I0 = 1; % Initial guess for I
vE0 = 0; % Initial guess for displacement error

% Define objective function
objfun = @(x) sum((P_hex - (48 * x(1) * E / L^3) .* (v_hex - x(2))).^2);

% Define objective function for bootstrapping
objfun_bootstrap = @(x,P_hex,E,L,v_hex) sum((P_hex - (48 * x(1) * E / L^3)
 .* (v_hex - x(2))).^2);

% Define constraints for I (I > 0)
options = optimoptions(@fmincon, 'Algorithm', 'interior-point');
lb = [0, -Inf]; % Lower bound (I > 0)
ub = [Inf, Inf]; % Upper bound

% Call fmincon to minimize objective function
x = fmincon(objfun, [I0, vE0], [], [], [], [], lb, ub, [], options);

% Extract optimal values
I_optimal = x(1);
vE_optimal = x(2);

% Define the number of bootstrap samples
numBootstraps = 1000;
bootstrap_estimates = zeros(numBootstraps, 2);

% Perform bootstrap estimation
n = length(P_hex);
for i = 1:numBootstraps
    % Generate bootstrap sample
    ind = randi(n, n, 1);
```

```
        P_b = P_hex(ind);
        v_b = v_hex(ind);

        % Call fmincon to minimize objective function for bootstrap sample
        x_b = fmincon(@(x) objfun_bootstrap(x, P_b, E, L, v_b), ...
                      [I0, vE0], [], [], [], [], lb, ub, [], options);
        bootstrap_estimates(i, :) = x_b;
    end

    % Compute confidence intervals
    alpha = 0.05; % 95% confidence interval
    I_conf_int = quantile(bootstrap_estimates(:,1), [alpha/2, 1-alpha/2]);
    vE_conf_int = quantile(bootstrap_estimates(:,2), [alpha/2, 1-alpha/2]);

    % Display the optimal value of I and vE
    disp(['Optimal value of I: ', num2str(I_optimal), ' mm^4']);
    disp(['Optimal value of vE: ', num2str(vE_optimal), ' mm']);

    % Display the confidence intervals
    disp(['95% confidence interval for I: [', num2str(I_conf_int(1)), ', ',
   num2str(I_conf_int(2)), '] mm^4']);
    disp(['95% confidence interval for vE: [', num2str(vE_conf_int(1)), ', ',
   num2str(vE_conf_int(2)), '] mm']);
end
```